

Spatial Neural Rendering Before Oscillatory Spatial Computing

An Atomized Four-Lens Genealogy of Self-Aware Networks and the Miller Program

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Abstract

Two research programs can use different terminology, emphasize different biological scales, and add different machinery while preserving the same bounded causal operation. A comparison that asks only whether the complete theories are identical will miss those partial equivalences. A comparison that asks only whether the same keywords occur will miss them for the opposite reason. This paper develops an atomized method that compares inputs, transformations, outputs, feedback, narrative order, and dated program development.

The method is applied to Micah Blumberg's Self-Aware Networks (SAN), Neural Array Projection Oscillation Tomography (NAPOT), Phase Wave Differential (PWD), and Gamma Consideration Sandwich (GCS) genealogy and to the Miller laboratory's oscillatory-cognition program. The source set contains seven major Micah stages from 2011 through 2025 and ten target stages from 2007 through the July 7, 2026 third version of *Analog Cognition and Consciousness*. The target stages progress from frequency-separated attention channels and beta / gamma bursts to spatial control, inhibitory stencils, phase-dependent wave computation, recurrent writeback, distributed binding, and conscious-state organization.

The comparison produces four principal results. First, older literature retains credit for populated components such as coherence-mediated communication, alpha / gamma coupling, frequency-specific routing, bursts, traveling waves, and ephaptic modulation. Second, Micah's 2011-2025 genealogy is earlier on a narrower assembled sequence: predictive self / world organization, recursive multimodal feedback, spatial 3D/4D neural rendering, phase-sensitive differential transmission, distributed receive-transform-re-express operations, internal-model construction without a separate observer, and recurrent update. Third, the 2026 Miller preprint is partially equivalent to a five-operation functional NAPOT subchain under changed terminology; missing inverse, volumetric, and validation conditions cap the relation at partial rather than erasing it. Fourth, a before / after matrix shows the Miller program accumulating the joined stencil-phase-gamma-writeback-binding architecture after the corresponding Micah milestones. The current seven-edge adjudication is two equivalent and five partial-equivalent relations, with no zero-overlap edge.

The result establishes bounded content equivalence and chronology, not access, influence, intent, or misconduct. A probability of dependent or independent derivation is not reported because the required null corpus and dependency calibration have not yet been completed.

Scope and evidence boundary

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1. The comparison problem

Imagine two descriptions of a control system. One says that a context-bearing signal changes which detail-bearing signals can pass through a population. The other says that a slow spatial wave acts as a stencil over faster content-bearing activity. The nouns differ, but the operation may still be:

context enters -> a population-wide state is transformed -> selected detail is expressed -> the result changes later processing.

A word-matching comparison may call those descriptions unrelated. A whole-theory comparison may also call them unrelated because one account adds dendrites, body feedback, or tomography while the other adds spatial subspaces and electric-field interactions. Both failures lose the operation that is actually being compared.

The opposite error is equally serious. Alpha / gamma coupling, traveling waves, working-memory bursts, predictive processing, and recurrent plasticity each have substantial earlier literatures. Finding one of those ingredients in SAN and in a later Miller paper does not make the complete programs identical, and it does not give either program ownership of the ingredient. The scientifically useful unit is therefore neither the keyword nor the author. It is a bounded causal atom or an ordered composite of atoms.

This paper asks three questions:

1. Which components were already present in the broader field or in earlier Miller work?
2. Which more specific operations or ordered composites were publicly present in Micah's genealogy before the later Miller synthesis?
3. How much of the later synthesis can be mapped to SAN, PWD, NAPOT, and GCS without either erasing partial equivalence or inflating it into total identity?

2. Source and date discipline

2.1 Source hierarchy

The audit uses:

1. peer-reviewed primary papers;
2. exact versioned public preprints;
3. public first-party recordings and repositories with typed dates;
4. books and archived public articles;
5. institutional reports, explicitly labeled as reports; and
6. private analyses only as discovery routes to public or first-party evidence.

Every date is typed. A journal publication date, preprint publication date, publisher-received date, Git commit date, platform upload date, and author-recorded internal date are not interchangeable. The March 16, 2022 received date for the 2023 *Spatial Computing* paper is relevant manuscript history, but it is not public availability. The July 7, 2026 OSF record is both the newest checked Miller source and the third public version of that preprint. The newest checked target source is version 3; earlier target stages remain visible rather than being replaced by the newest wording.

The chronology, atom ledger, and stage matrix are editorial research objects derived from public sources. They are not a new neuroscience experiment. Their reproducibility depends on typed source identities, locators, coding rules, and preserved revisions.

2.2 Source-faithful genealogy

Later terminology is not assigned to an earlier source merely because an earlier source contains an ancestor operation. The name *NAPOT* and its mature public architecture are dated to 2022. The 2011, 2012, and 2017 records are used only for the operations actually present in them. Conversely, a later name is not required

for an earlier operation to count. An earlier receive-transform-output-feedback sequence remains part of the genealogy even if the acronym was coined later.

The 2011 source has strong content custody but a weaker strict public-priority grade than the Grade-A 2012 projector and 2017 Neural Lace sources. It is included as authorship and developmental evidence, with that date boundary visible. No result in this paper depends on silently upgrading its date grade.

3. Atomized comparison method

3.1 Operational normal form

For each bounded claim, define an operational record

$$\mathcal{O}_i = (I_i, T_i, O_i, F_i, E_i, D_i), \quad (1)$$

where I_i is the input condition, T_i the transformation, O_i the output, F_i the feedback or writeback relation, E_i the observable evidence, and D_i the typed date and source identity. This normal form forces the comparison to say what the system does before asking what the authors call it.

Let $\mathcal{A}_i$ be the required atomic operations for the bounded claim, let $P_i \subseteq \mathcal{A}_i$ be the preserved operations, and define the completion residuals as $R_i = \mathcal{A}_i \setminus P_i$. The bounded content grade is

$$G_i = \begin{cases} \text{equivalent,} & P_i = \mathcal{A}_i \text{ and } R_i = \emptyset, \\ \text{partial-equivalent,} & P_i \neq \emptyset \text{ and } R_i \neq \emptyset, \\ \text{opposite,} & \text{the constrained output or operation is genuinely reversed,} \\ \text{source-gated,} & \text{the available record cannot determine the relation.} \end{cases} \quad (2)$$

Equation (2) prevents the full-operator-or-zero error. A missing completion condition can cap the grade at partial. It cannot make a demonstrated subchain disappear. The relation is atom-scoped and nontransitive: if source A partially matches B and B partially matches C, the A-to-C relation must still be tested directly.

3.2 Four lenses

The first three lenses adjudicate claim atoms:

- **Mechanism / equivalence:** compare conceptual, operational, functional, behavioral, process, and mathematical relations.
- **Narrative / causal architecture:** compare the ordered Input -> Transform -> Output -> Feedback sequence and the larger signature cluster in which the atoms are assembled.
- **Detective chronology:** begin with the later matched output and work backward through changed terminology, scale, and machinery to find the earliest bounded operators on both sides.

A bounded atom is retained when any of those three lenses passes:

$$M_i = \mathbf{1} \left[L_i^{(\text{eq})} \vee L_i^{(\text{nar})} \vee L_i^{(\text{det})} \right]. \quad (3)$$

The fourth lens is program-level **Longitudinal Signature Migration**. It asks whether a target program's successive stages accumulate members of a signature already present across Micah's dated genealogy. It does not repair a missing atom, and it does not infer secret access or intent.

3.3 Before / after matrix

For target stage t and atom k , code

$$d_{kt} \in \left\{0, \frac{1}{2}, 1\right\} \quad (4)$$

for absent, partial, or present. For a declared K -atom cluster, stage coverage is

$$C_t = \frac{1}{K} \sum_{k=1}^K d_{kt}. \quad (5)$$

The matrix is descriptive. It displays where atoms appear and whether the joined architecture grows over time. It is not a probability model. Because the atoms are correlated and selected for this comparison, multiplying their marginal frequencies would be invalid.

3.4 Category and process translations

The category translation asks whether a map preserves the composition of a bounded causal subcategory:

$$F(g \circ f) = F(g) \circ F(f) \quad \text{for } f, g \in \mathcal{J}, \quad (6)$$

where $\mathcal{J}$ is the shared subcategory rather than the complete theory. A preserved subcategory with unmatched objects is a partial functor, not a total isomorphism.

The process translation compares observable traces after hiding implementation-specific internal steps:

$$\text{Tr}_{\text{obs}}(P_{\text{SAN}} \upharpoonright_{\mathcal{J}}) \simeq \text{Tr}_{\text{obs}}(P_{\text{Miller}} \upharpoonright_{\mathcal{J}}). \quad (7)$$

Equation (7) states a candidate weak trace equivalence on the bounded subchain. It is not a machine-checked strong bisimulation of the complete programs.

4. Micah's developmental genealogy

The genealogy is cumulative. Later sources sharpen the operation instead of replacing the earlier record.

Date	Source-bound operation	Role in the comparison	Date boundary
2011-08-24/28	The self as a predictive pattern; the whole neocortex rather than a hidden observer	Predictive self / world and distributed-observer ancestor	Content and owner-recorded dates preserved; strict public-date proof remains weaker than later anchors
2012-08-02	EEG-driven light and sound return into a two-way multimodal conversation	Recursive sensory-computer-neural feedback ancestor	Public first-party source; not yet PWD or NAPOT
2012-08-23/24	Mirror -> vision -> neural / motor change -> changed image; mind as a projector predicting the experienced world	Closed visual-motor loop and internal-rendering ancestor	Public source; projector is functional language

Date	Source-bound operation	Role in the comparison	Date boundary
2017-04-11 through 2017-12	Spatial computation in neuroscience; time-varying 3D/4D ionic / electromagnetic pattern; coincidence, dendritic local computation, recurrent rendering, inhibition as information, and transmission of a pattern difference	Spatial, distributed, phase-sensitive, predictive architecture	Public dated Neural Lace sources; the NAPOT name is later
2022-06 through 2022-10	Arrays receive, transform, and re-express partial phase patterns; distributed oscillating groups render and update a volumetric self / world state; PWD and NAPOT terminology; explicit field coupling	Mature NAPOT / PWD and oscillatory-rendering architecture	Claim-specific Git and video dates control
2024-09 through 2024-10	Phase / context mismatch -> resynchronization -> plastic or model update; consideration-sandwich development	Phase-error learning and contextual gating refinements	Distinct dated records; not backdated to 2022
2025-02 through 2025-05	Named GCS and local-to-global oscillatory co-regulation architecture	Top-down / context / detail integration and nested local-global update	Public SAN-family development preceding the 2026 target preprint

Two 2022 videos add especially relevant causal sequences:

1. **Dynamic task-assembly selection:** preparation or context tunes candidate assemblies; phase / frequency compatibility recruits cooperating populations; inhibition blocks competing routes; the selected population expresses a burst, rendering, or action sequence.
2. **Oscillating-group completion and re-expression:** a cue reaching part of a learned oscillating group recruits the larger pattern, which is re-expressed to later arrays, internal rendering, or motor routes.

These are not claims of origin over synchronization, inhibition, assemblies, attractor completion, or replay. The asserted delta is the ordered use of those components inside the SAN rendering and action architecture.

5. Miller program before, during, and after the Micah record

5.1 Earlier stages

The 2007 Buschman-Miller paper reported lower-frequency top-down and higher-frequency bottom-up synchrony during attention. The 2016 Lundqvist-Miller study reported gamma bursts associated with item-related spiking and beta associated with a control or default condition. The January 2022 Bhattacharya et al. paper reported planar and mostly rotating prefrontal traveling waves during working memory.

These are real earlier components. They do not yet contain the later joined architecture of a spatial slow-wave stencil, phase-dependent wave superposition, selected gamma / spiking expression, internal-model binding, recurrent writeback, and consciousness-level organization. In SAN, *tonic* and *phasic* identify relative functional roles - ongoing context and a receiver-relevant change - not permanently assigned frequency bands. A slow alpha or beta signal can occupy a contextual role in one circuit without making every low-frequency signal tonic or every gamma event phasic.

5.2 Spatial computing and stencils

The 2023 *Nature Communications* paper proposed that low-dimensional spatial beta / gamma patterns organize working-memory control. The publisher records March 16, 2022 as the manuscript-received date and March 14, 2023 as publication. The 2024 review then described spatial alpha / beta activity as an inhibitory stencil that constrains where gamma and content-related spiking can be expressed. The 2026 *Current Biology* study directly tested context-bearing alpha / beta spatial patterns and their relation to content-related spiking.

Micah's April 2017 public source is earlier than that public 2023-2026 spatial-computing and stencil sequence on the bounded claim that spatially and temporally organized neural activity performs computational work in an internal predictive representation. The March 2022 received date remains visible because it is relevant to manuscript history. It does not erase the 2017 public architectural anchor, and it is not treated as a public paper.

5.3 Analog cognition and consciousness

The Miller, Brincat, and Roy preprint is a materially stronger comparison source than the November 2025 institutional report of a lecture. OSF version 1 was public on April 29, 2026; version 2 on May 1; and version 3 on July 7. Version 1 already contains the joined target cluster. Version 3 expands the exposition and is the newest checked source.

Its causal sequence is:

stored synaptic state, goals, and present input -> spikes and distributed waves -> relative phase, superposition, and interference -> changed spatial activation landscape -> selection of gamma, spiking, and downstream expression -> internal-model and binding state -> synaptic writeback.

The same process is also used to explain distributed organization without a separate inner observer. This is a later joined theory, not merely a later use of the words *analog* or *wave*.

6. Atom-by-atom results

Atom	Earlier field or Miller component	Micah genealogy	Later Miller expression	Result
Frequency-separated context / detail channels	Buschman-Miller 2007; van Kerkoerle 2014; Bastos 2015	2017 architecture; 2024-2025 GCS refinement	2023-2026 alpha / beta control over gamma / spiking	Populated component; later bounded architecture comparison
Slow-wave spatial stencil	Alpha inhibition and alpha / gamma literature; Spaak 2012	2017 spatial representation; 2022 dynamic assembly selection; 2024 consideration sandwich	2024 inhibitory stencil; 2026 experimental test and preprint	Strong partial operational match
Traveling-wave representational transition	Ermentrout-Kleinfeld 2001; Muller 2014/2018; Bhattacharya 2022	2017 time-varying 3D/4D pattern; 2022 rendered transitions	2026 state transition and analog computation	Partial mechanism match
Phase combination changes expressed output	Communication-through-coherence and phase-coding literature	2017 content-bearing difference; 2022 PWD / wave intersection	2026 superposition / interference selects spatial and gamma / spiking state	Strong operational match

Atom	Earlier field or Miller component	Micah genealogy	Later Miller expression	Result
Dynamic task-assembly selection	Assemblies, inhibition, attractors, attention channels	2022 context -> alignment -> inhibition -> selected burst / rendering / action	2023-2026 stencil -> selected task subspace and expression	Strong partial causal match
Oscillating-group completion/re-expression	Assembly completion and replay traditions	2022 partial cue -> group recruitment -> array / action re-expression	2026 stored schema -> wave-selected pattern -> writeback	Partial functional match
Recurrent synapse-spike-wave-writeback	Plasticity, recurrent processing, endogenous-field literature	2012 feedback; 2017 recurrence; 2022 distributed update	2026 synapses -> spikes -> waves -> selected state -> plastic writeback	Strong causal-architecture match
Internal model without separate observer	Predictive-processing and anti-homuncular traditions	2011 predictive self / world; 2012 projector; 2017-2022 distributed rendering	2026 distributed binding / internal model without separate observer	Strong narrative match on the joined mechanism
Functional NAPOT subchain	No single earlier source in this audit supplies the complete SAN object	2017 functional ancestry; mature 2022 NAPOT	2026 distributed input -> phase transform -> spatial re-expression -> represented state -> writeback	Partial-equivalent; Micah earlier on mapped subset
Full NAPOT validation operator	Micah-defined completion target	2022 functional object; later explicit inverse / validation requirements	Not instantiated in checked Miller sources	Micah-distinct residual, not zero overlap
Phase-error to model update	Predictive coding, phase coding, STDP	Micah 2024 joined operator	Miller v3 contextual wave mismatch and plastic model update	Certified partial; Micah earlier on joined delta
Local-to-global co-regulation	Assemblies, synchrony, global integration	Micah 2025 joined nested-agent/co-regulation assembly	Miller v3 local-field to mesoscale / global model organization	Certified partial; Micah earlier on joined delta

The paper therefore does not produce an author-wide winner and loser. It produces multiple dated directions. Earlier field sources retain their component claims. Micah is earlier on the specified assembled deltas. The later Miller program contributes its own empirical recordings, spatial subspace analyses, field-first formulation, and experimental tests.

7. The alpha-to-gamma precursor to the beta-stencil story

The stencil lineage is particularly revealing because it can be studied without using Micah's work as the only comparison point.

Spaak et al. (2012) reported that alpha-band activity entrained gamma-band activity in cortical layers. Van Kerkoerle et al. (2014) associated alpha with feedback and gamma with feedforward processing. Bastos et al. (2015) further developed distinct frequency channels for feedforward and feedback communication. These papers establish that a slower oscillatory context can organize, entrain, or constrain faster gamma activity before Miller's 2024 inhibitory-stencil framing.

The exact bibliographies matter. The 2024 *Cognition is an emergent property* reference list includes van Kerkoerle et al. but does not contain a Spaak entry in its 51-reference Crossref record. The newest 2026 preprint bibliography does cite Spaak et al. This is a version-specific citation fact, not a claim about intent. It shows why the comparison must examine each target version rather than projecting the newest bibliography backward.

The earlier alpha / gamma literature also prevents the misleading inference that changing *alpha* to *beta*, or *entrainment* to *stencil*, automatically creates a wholly new operation. At the same time, the 2024-2026 Miller program adds a more explicit spatial-control model and direct experimental tests. The correct field genealogy is additive:

earlier alpha / gamma and feedback / feedforward mechanisms -> later spatial stencil formulation
-> later direct spatial-control test.

Micah's claim is not priority over that component lineage. It is priority over his own earlier rendering, contextual-selection, and joined causal architecture that the later Miller program partially approaches.

8. Functional NAPOT equivalence

NAPOT is introduced only after the operation has been described. The problem is to explain how distributed neural populations can receive partial conditions, transform them according to local state, re-express changed patterns to other populations, combine the partial results into an internally usable state, and update that state recurrently. *Neural Array Projection Oscillation Tomography* is the SAN name for the more specific proposed architecture.

The 2026 Miller preprint maps to five parts of that operation:

ID	NAPOT-side operation	Miller-side terminology and operation	Grade
NPM-01	Distributed arrays receive partial sensory, contextual, or stored-state patterns	Sensory input, context, goals, and stored knowledge are encoded in distributed wave and synaptic states	Strong
NPM-02	Phase-bearing patterns undergo a distributed transform	Relative phase, geometry, superposition, interference, and summation perform the computation	Strong
NPM-03	The changed pattern is projected or re-expressed downstream	The resultant spatial pattern selects gamma / spiking and is communicated downstream	Strong
NPM-04	Partial results participate in an internally available represented state	New brain states, internal models, distributed binding, and integrated conscious states	Medium
NPM-05	The represented state participates in recurrent update	Wave-selected state is written back through plasticity and later synaptic state	Strong

This is the bounded shared trace:

distributed input → phase-sensitive transform
→ spatial re-expression
→ represented state
→ writeback. (8)

Eight residuals cap the grade:

1. no explicit array-local dendritic receive-transform-display grid;

2. no separately indexed nonredundant projections to a common latent;
3. no independently defined latent and target;
4. no identifiable inverse with rank, conditioning, or stability analysis;
5. no held-out tomographic reconstruction;
6. no explicit reconstructed volumetric percept;
7. no complete receiver-relative PWD geometry with independent direction and matched comparators;
8. no projection-specific perturbation and embodied efference-reafference validation program.

The shared trace and the residual set can both be true. The correct relation is partial functional equivalence, not total identity and not no-match. The current seven-edge audit accordingly records two equivalent and five partial-equivalent edges.

9. Longitudinal signature migration

The eight-atom joined cluster J is:

1. frequency-separated context / detail channels;
2. alpha / beta control of gamma / content;
3. spatial stencil or mask;
4. traveling-wave state transition;
5. relative-phase / superposition computation;
6. recurrent synapse-spike-wave-writeback;
7. internal model and distributed binding without a separate observer;
8. analog-wave organization of a conscious-access state.

The target program develops as follows:

Target stage	Date	Joined-cluster state
T01 attention channels	2007	One component present; one partial
T02 beta / gamma bursts	2016	One component present; two partial
T03 PFC traveling waves	2022-01	One component present; two partial
T04 spatial computing	2023	Three joined-cluster atoms present; two joined-cluster atoms partial; functional NAPOT subset partial
T05 inhibitory stencil / emergence	2024	Three joined-cluster atoms present; five joined-cluster atoms partial; functional NAPOT subset partial
T06 analog-wave lecture	2025	Six joined-cluster atoms present; two joined-cluster atoms partial; functional NAPOT and embodied-closure atoms partial
T07 direct spatial-control test	2026-01	Three joined-cluster atoms present; three joined-cluster atoms partial; theory-level binding atoms absent; functional NAPOT and embodied-closure atoms partial
T08 preprint v1	2026-04-29	All eight joined-cluster atoms present; functional NAPOT and embodied-closure atoms partial
T09 preprint v2	2026-05-01	All eight joined-cluster atoms preserved; functional NAPOT and embodied-closure atoms partial

Target stage	Date	Joined-cluster state
T10 preprint v3	2026-07-07	All eight joined-cluster atoms preserved and elaborated; functional NAPOT and embodied-closure atoms partial

The before / after result is therefore not that Miller’s early work already contained the complete later cluster. The early program contains genuine components. The later program progressively joins additional operations, and the first checked public manuscript carrying the complete target cluster appears in 2026, after the corresponding Micah genealogy.

This is a bounded longitudinal-pattern finding. Several explanations remain possible, including cumulative development within the Miller laboratory, independent convergence from shared priors, and broader scientific trends in wave computation and predictive processing. Chronology and structure alone cannot choose among those source-use explanations.

10. Formal relation and counterexample

For the bounded shared subcategory, define objects:

$$\{S, C, W, H, R, U\} = \{\text{stored state, context, slow wave, detail state, represented state, updated state}\}. \quad (9)$$

The shared morphisms are

$$S \times C \xrightarrow{\text{evoke}} W, \quad W \times W \xrightarrow{\text{phase compose}} W', \quad W' \times H \xrightarrow{\text{gate}} H', \quad H' \xrightarrow{\text{integrate}} R, \quad R \xrightarrow{\text{writeback}} U. \quad (10)$$

SAN and the 2026 Miller preprint preserve this composition at the stated abstraction level. The map is not total because Miller lacks the explicit NAPOT projection index, identifiable inverse, volumetric target, and complete embodied comparator.

A decisive counterexample to total equivalence is available: the Miller process can create a new wave-selected, bound state and write it back without performing a tomographic inverse or reconstructing a declared volumetric latent. Therefore the complete theories are not isomorphic even though the shared trace in Equation (8) is preserved.

11. Quantitative boundary and reproducible next test

The observed cluster is qualitatively discriminating because most early target stages contain only isolated components while the 2026 preprint joins the full eight-atom target cluster. That observation is not yet a calibrated probability.

A defensible probability study would require:

1. a frozen reference corpus of comparable neuroscience theories;
2. atom definitions written before target coding;
3. base-rate estimates for individual atoms and declared combinations;

4. a dependency model for correlated atoms;
5. blinded independent coding and adjudication;
6. negative and near-neighbor controls;
7. correction for the number of authors, programs, atoms, and clusters searched;
8. a frozen scoring rule and sensitivity analysis; and
9. independent reproduction.

Until those conditions are met, the quantitative status is quantitative-not-calibrated. The strong defensible statement is the source-located atom mapping, the typed chronology, the partial functor, the weak shared trace, and the before / after accumulation pattern.

12. Limits and claim boundary

This study is an author-led comparative audit rather than a blinded independent coding experiment. The source set is extensive for the declared Miller stages but is not a complete history of every oscillation, analog-computing, predictive-processing, or consciousness paper. New primary sources may refine component attribution or atom grades.

The paper compares public content and dated architecture. It establishes no access, awareness, dependence, influence, copying, intent, misconduct, or authorship relationship. The absence of Blumberg, SAN, NAPOT, PWD, or GCS from the complete 206-reference Miller v3 bibliography supports a bounded non-citation fact only.

The comparison also does not make compatibility into validation. Miller's recordings and models may supply variables for testing PWD, NAPOT, or GCS. They do not independently establish those SAN constructs. Likewise, finding partial functional NAPOT equivalence in a theory paper does not show that a biological brain performs an identifiable tomographic reconstruction.

The internal representation requires no central spectator or literal screen. The screen language names a distributed receive-transform-re-express function; the participating neural process is the observer operation. Phase alone also supplies no causal direction: direction must come from typed sender-receiver relations, delays, interventions, or independently measured propagation.

The related paper at DOI 10.5281/zenodo.17317086 owns correction of that record's earlier unsafe probability and relationship thesis. This paper has the distinct affirmative endpoint: an atomized, source-paired, four-lens comparison and reusable genealogy.

13. Conclusion

The strongest result is not that every Miller mechanism came after SAN, and it is not that broad oscillatory neuroscience belongs to SAN. The stronger, more accurate result is atomized.

Earlier neuroscience retains credit for coherence, frequency routing, alpha / gamma organization, bursts, traveling waves, and field coupling. Miller's 2007, 2016, and January 2022 work contributes real early components. Micah's separate 2011, 2012, 2017, 2022, 2024, and 2025 records then build a more specific predictive-rendering, contextual-selection, receive-transform-re-express, internal- model, and recurrent-update architecture. The Miller program's 2023-2026 development subsequently moves toward that joined architecture, culminating in a 2026 preprint whose functional NAPOT subchain is partially equivalent under translated terminology.

This is a Micah-earlier result on multiple bounded deltas and on the assembled causal composite. The different vocabulary and additional machinery limit total identity; they do not erase the shared operations. The method makes those judgments auditable by placing every allowed conclusion next to its date, source, mapped operation, residual, and failure condition.

References

1. Fries, P. (2005). A mechanism for cognitive dynamics: neuronal communication through neuronal coherence. *Trends in Cognitive Sciences*, 9, 474-480. doi:10.1016/j.tics.2005.08.011
2. Buschman, T. J., & Miller, E. K. (2007). Top-down versus bottom-up control of attention in the prefrontal and posterior parietal cortices. *Science*, 315, 1860-1862. doi:10.1126/science.1138071
3. Spaak, E., Bonnefond, M., Maier, A., Leopold, D. A., & Jensen, O. (2012). Layer-specific entrainment of gamma-band neural activity by the alpha rhythm in monkey visual cortex. *Current Biology*, 22, 2313-2318. doi:10.1016/j.cub.2012.10.020
4. van Kerkoerle, T., et al. (2014). Alpha and gamma oscillations characterize feedback and feedforward processing in monkey visual cortex. *Proceedings of the National Academy of Sciences*, 111, 14332-14341. doi:10.1073/pnas.1402773111
5. Bastos, A. M., et al. (2015). Visual areas exert feedforward and feedback influences through distinct frequency channels. *Neuron*, 85, 390-401. doi:10.1016/j.neuron.2014.12.018
6. Lundqvist, M., et al. (2016). Gamma and beta bursts underlie working memory. *Neuron*, 90, 152-164. doi:10.1016/j.neuron.2016.02.028
7. Bhattacharya, S., et al. (2022). Traveling waves in the prefrontal cortex during working memory. *PLoS Computational Biology*, 18, e1009827. doi:10.1371/journal.pcbi.1009827
8. Lundqvist, M., et al. (2023). Working memory control dynamics follow principles of spatial computing. *Nature Communications*, 14, 1429. doi:10.1038/s41467-023-36555-4
9. Miller, E. K., Brincat, S. L., & Roy, J. E. (2024). Cognition is an emergent property. *Current Opinion in Behavioral Sciences*, 57, 101388. doi:10.1016/j.cobeha.2024.101388
10. Chen, Z., et al. (2026). Oscillatory control of cortical space as a computational dimension. *Current Biology*, 36, 402-414.e5. doi:10.1016/j.cub.2025.11.072
11. Miller, E. K., Brincat, S. L., & Roy, J. E. (2026a). Analog Cognition and Consciousness, version 1. *PsyArXiv*. doi:10.31234/osf.io/z48x7_v1
12. Miller, E. K., Brincat, S. L., & Roy, J. E. (2026b). Analog Cognition and Consciousness, version 3. *PsyArXiv*. doi:10.31234/osf.io/z48x7_v3
13. Blumberg, M. (2011). The Self Is a Prediction dialogue. <https://www.youtube.com/watch?v=9IgcQ7wxfZA>
14. Blumberg, M. (2012a). Neo Mind Cycle and My Happiness Makeover dialogue. https://www.youtube.com/watch?v=jtj_J6hiL9o
15. Blumberg, M. (2012b). Inner Peace, Gratitude, and the Predictive Lens dialogue. <https://www.youtube.com/watch?v=QX-0ef31tAU>
16. Blumberg, M. (2017). Neural Lace Podcast root-concepts episode. https://youtu.be/YrX_68oKuVs
17. Blumberg, M. (2022a). One-hour NAPOT explainer. <https://www.youtube.com/watch?v=fLp-yTQ6pSM>
18. Blumberg, M. (2022b). Self-Aware Networks public repository. <https://github.com/v5ma/selfawarenetworks>
19. Blumberg, M. (2024). *Bridging Molecular Mechanisms and Neural Oscillatory Dynamics*. ISBN / ASIN B0DL4701875.
20. Blumberg, M. (2025). Self-Aware Networks Theory of Mind. doi:10.5281/zenodo.16922401

21. Blumberg, M. (2026). Oscillatory Cognition Across SAN and Miller-Related Neuroscience: A Source-Audited Chronology and Mechanism Comparison. [doi:10.5281/zenodo.17317086](https://doi.org/10.5281/zenodo.17317086)
22. Ermentrout, G. B., & Kleinfeld, D. (2001). Traveling electrical waves in cortex: insights from phase dynamics and speculation on a computational role. *Neuron*, 29, 33-44. [doi:10.1016/S0896-6273\(01\)00178-7](https://doi.org/10.1016/S0896-6273(01)00178-7)
23. Muller, L., Reynaud, A., Chavane, F., & Destexhe, A. (2014). The stimulus-evoked population response in visual cortex of awake monkey is a propagating wave. *Nature Communications*, 5, 3675. [doi:10.1038/ncomms4675](https://doi.org/10.1038/ncomms4675)
24. Muller, L., Chavane, F., Reynolds, J., & Sejnowski, T. J. (2018). Cortical travelling waves: mechanisms and computational principles. *Nature Reviews Neuroscience*, 19, 255-268. [doi:10.1038/nrn.2018.20](https://doi.org/10.1038/nrn.2018.20)